BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: KT-IX-23-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** Code B

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. Conclusion of Ruther's Ford experiment is discovery of .....

(A) Electron
(B) proton
(C) Nucleus
(D) Neutron

Qii. Element "X" has valency +2, in which group it is placed ?.

(A) Alkali metals
(B) Alkaline earth metals
(C) Boron family
(D) Halogens

Qiii. A chemical bond formation that involves the complete transfer of electrons between atom is ...

(A) Ionic bond
(B) Covalent bond
(C) Metallic bond
(D) partial Covalent bond

Qiv. If electro negativity difference is less than 1.7, then bond will be .....

(A) Ionic bond
(B) Covalent bond
(C) Co-ordinate Covalent bond
(D) Metallic bond

Qv. When a nail strucks in tyres of a car the tyre collapse due to ...

(A) Diffusion
(B) Effusion
(C) compressibility
(D) atmospheric pressure

Qvi. What is Molarity of solution when 58.5g of NaCl is dissolved in 200ml of water?.

(A) 0.2 M
(B) 0.5 M
(C) 0.05 M
(D) 0.02 M

Qvii. In solution the particles are ……

(A) Invisible
(B) Visible by naked eye
(C) Visible by ordinary scope
(D) Visible by electron microscope

Qviii. Which of the following is the most powerful oxidizing agent?.

(A) F_2
(B) Cl_2
(C) Br_2
(D) I_2

Qix. What is the oxidation number of nitrogen in NO_2^{-1} ?.

(A) +1
(B) +2
(C) +3
(D) +4

Qx. Scientists redefined the metal on the basis of ……

(A) Loss and gain of electron
(B) Loss of electron
(C) gain of electron
(D) No change in electron number

Qxi. Dual nature of matter is determined by ……

(A) Rutherford
(B) Neil Bohr
(C) De-Broglie
(D) J.J Thomson

Qxii. Which of the following is considered as cation?.

(A) Na
(B) Cl^-
(C) Mg^{+2}
(D) CH^+_3

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

QQ.2. Attempt any EIGHT parts. All parts carry equal marks.

i How many moles are present in 150g of $CaCO_3$? (At. mass Ca = 40amu, C = 12 amu, O = 16 amu)

ii Find the number of proton and neutron for (a). Nitrogen Z.= 7 , A = 14 (b). Aluminum, Z = 13 , A = 27

iii Identify the group and periods with electronic configuration as follows: (a). $1s^2$, $2s^2$, $2p^6$, $3s^1$. (b). $1s^2$, $2s^2$, $2p^1$, $3s^2$, $1s^2$, $3p^6$, $4s^2$

iv Why element helium and Argon are chemically inactive?. Support your answer with the help of their electronic configuration.

v With the help of dot cross model show the formation of: (a). Single covalent bond between Chlorine atom. (b). Double covalent bond between Oxygen atoms.

vi A container of gas has a volume of 1500ml at 22 °C. What volume will the gas occupy at 0 °C. If pressure of gas remains constant?.

vii Identify solute and Solvent in carbonated drink and Jewelry.

viii How many ml of solvent are required to make 0.2 Mole solution in 500 ml from a stock solution of 2 mole.

ix Write chemical reactions occurring in Dawn Cell.

x What does the term cathodic protection means. Give some examples.

xi a. Why Nobel metals are considered as precious and valuable metals?. b. What do you mean by word Nobel metals?.

Section "C"

Maximum Marks: 21

> Note: Note: Attempt any THREE of the following questions. All questions carry equal marks.

QQ.3. (Marks 3,4)

a [3] Find out molecular mass of $C_6H_{12}O_6$ in a.m.u and express it into grams?.

b [4] Draw the atomic structure of: $^{12}_{6}C$, $^{35}_{17}Cl$, $^{234}_{92}U$ and $^{13}_{6}C$

QQ.4. (Marks 3,4)

a [3] What is meant by groups and periods in periodic table?.

b [4] What are the main types of bonds?. Describe how each is formed?.

QQ.5. (Marks 3,4)

a [3] State mathematically the effect of pressure on the volume of a gas?.

b [4] Define solute and solvent. Elaborate your answer with the help of suitable examples?.

QQ.6. (Marks 3,4)

a [3] Why sodium is considered as alkali metal?. What position is occupied by sodium in periodic table?.

b [4] How electrical energy is produced in voltic cell. Write the composition of electrodes in Daniel cell.